

EfficientNetB0-Based Deep Transfer Learning Framework for Accurate Medicinal Plant Identification

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Abstract—Medicinal plants play an important role in both traditional and modern medicine due to their inherent medicinal properties. It is important to correctly identify medicinal plants in order to avoid negative reactions. However, medicinal plant identification is challenging due to similarities in leaves, effects of the environment, and the need for botanical knowledge. This paper proposes an automatic system for the identification of medicinal plant leaves using deep transfer learning. This system uses the pre-trained EfficientNetB0 model which is a lightweight convolutional neural network trained on the ImageNet dataset in identifying the unique characteristics of leaves. The system also requires preprocessing, resizing, and augmenting data to make the model more reliable. The experiments were carried out on a dataset containing nearly 6,000 leaf images of 40 medicinal plants. It showed that the model was able to achieve an accuracy rate greater than 90% with less training time and lower computational power required. These results indicate that EfficientNetB0 could be a useful tool for real-time identification of medicinal plants and can thus be incorporated into mobile and embedded health systems.

Index Terms—Medicinal Plant Identification, Deep Learning, Transfer Learning, EfficientNetB0, Image Classification

I. INTRODUCTION

Medicinal plants play an important role in both the traditional and modern practice of medicine since they have healing properties and comprise several active chemicals. Ayurvedic medicine, Siddha medicine, Unani medicine, and Chinese medicine all depend on plant medicines. A large number of people around the world still take herbal medicines, especially in developing nations [1]. Thus, accurate identification of medicinal plant species is essential for their safe use and proper treatment.

The identification of medicinal plants manually is not feasible as many plant species possess similar leaf shapes, textures, and vein structures. The traditional method of plant identification relies on expertise of the botanist, thus making it an expensive and time-consuming process that is not suitable for large-scale operations [2]. Early efforts towards automation of plant recognition utilized hand-crafted features and machine learning classifiers such as SVM and k-NN [3]. This approach was not efficient due to the sensitivity of algorithms to changes in environmental conditions, particularly variations in lighting and background noise.

However, recent developments in the domain of deep learning have revolutionized image recognition tasks by enabling automated extraction of essential features from images. In particular, CNN-based deep models such as VGGNet and ResNet have demonstrated excellent results in plant and leaf recognition tasks. However, training these deep models from the ground up needs large datasets and significant computing power, which are often lacking in medicinal plant research [4], [5].

Transfer learning provides an effective solution to these challenges by adapting pretrained models to domain-specific classification tasks. Among the latest CNN architectures, EfficientNet has received a lot of attention for its compound scaling strategy. This approach achieves high accuracy while using fewer parameters [6]. Specifically, EfficientNetB0 strikes a good balance between computational efficiency and recognition accuracy. This makes it a great fit for real-time identification of medicinal plant leaves. In this paper, we propose a deep transfer learning framework based on EfficientNetB0 for

accurate and efficient classification of medicinal plant leaves.

II. LITERATURE SURVEY

Many researchers have tried to automate the process of plant and medicinal leaf recognition using image processing and machine learning. In the initial phase, most research papers were based on the extraction of handcrafted features and traditional machine learning classifiers. For example, Kumar et al. [1] used texture and shape features along with traditional machine learning classifiers and showed moderate results. However, these systems were highly prone to environmental changes such as illumination and complex backgrounds.

With the evolution of deep learning, Convolutional Neural Networks (CNNs) have been used extensively for plant recognition tasks. Singh et al. [2] used deep CNN models for leaf recognition and showed substantial improvement over handcrafted feature-based approaches. Similarly, Sharma and Gupta [3] showed the efficacy of transfer learning using pre-trained CNN models for medicinal plant classification with improved accuracy and less training time.

Architectures like VGGNet, proposed by Simonyan and Zisserman [4], which are deep CNN architectures, have also been used in leaf classification tasks, resulting in better recognition accuracy but at the expense of increased computational complexity. Additionally, residual learning-based architectures like ResNet, proposed by He et al. [5], successfully tackled the degradation problem in deep neural networks and helped in better feature extraction for plant recognition tasks.

Recently, efficient architectures have emerged as promising candidates in medicinal plant recognition tasks. EfficientNet, proposed by Tan and Le [6], introduced a compound scaling method that resulted in high accuracy with reduced parameters, making it an ideal candidate for real-time plant leaf recognition systems. The groundbreaking CNN model proposed by Krizhevsky et al. [7] also paved the way for using deep learning in image-based classification tasks and motivated further research in plant recognition. The comparison of previous studies is presented in Table I.

III. PROPOSED METHODOLOGY

This section describes the entire process of the proposed deep transfer learning framework using EfficientNetB0 for the identification of medicinal plant leaves. The proposed framework includes the following: dataset preparation, preprocessing, augmentation, model design, training, and evaluation.

A. System Overview

The proposed framework is expected to automate the identification of medicinal plant species based on leaf image data using deep convolutional feature learning. The end-to-end process involves dataset collection, followed by preprocessing and augmentation to enhance the quality and variability of the images. The EfficientNetB0 model is utilized as the main feature extractor via transfer learning, and a custom classification head is trained to make predictions on plant species. The trained model is assessed based on accuracy, precision, recall, F1-score, and confusion matrix analysis as shown in fig.1.

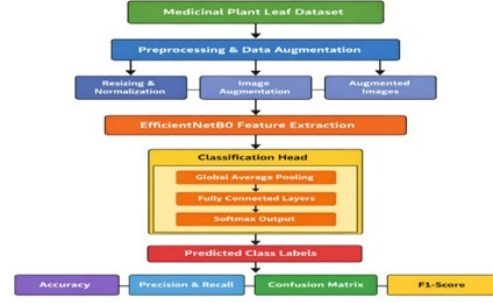


Fig. 1. Overall architecture of the proposed EfficientNetB0-based framework

B. Dataset Description

The experiments were carried out using a medicinal plant leaf image dataset that comprises around 6,000 images of 40 different species of plants. The dataset comprises leaf images collected in different environmental conditions, including varying lighting conditions, complex backgrounds, and varying leaf orientations. This is a challenging task for classification because of the high similarity between the leaves of different plant species. To test the performance of the model without any bias, the dataset was split into training, validation, and testing sets in the ratio of 80:10:10 as shown in fig.2.



Fig. 2. Sample medicinal plant leaf images from the dataset

C. Image Preprocessing

Before the training process, all leaf images were normalized to conform to the input requirements of the EfficientNetB0 model. The images were all resized to 224×224 pixels. Normalization of the pixels was also done to scale the intensity values of the pixels. This helps in the convergence process during training. The mathematical expression for the normalization process is given by:

$$I_{norm} = \frac{I - \mu}{\sigma} \quad (1)$$

where I represents the original image, μ is the mean pixel value, and σ denotes the standard deviation. The image as shown in fig.3.

TABLE I
LITERATURE SURVEY ON MEDICINAL PLANT LEAF IDENTIFICATION USING DEEP LEARNING

Ref.	Study Title / Focus	Dataset/Source	Preprocessing Methods	Model Used	Key Findings
[1]	Medicinal plant identification using leaf images	Medicinal leaf dataset	Feature extraction, normalization	Traditional ML + hand-crafted features	Performance affected by illumination and background complexity.
[2]	Plant leaf classification with deep CNNs	Leaf image benchmark datasets	Resize, augmentation	Deep CNN architectures	Higher recognition accuracy compared with conventional approaches.
[3]	Transfer learning for medicinal plant recognition	Self-collected medicinal plant leaves	Data augmentation, re-sizing	Pretrained CNN models	Reduced training time with higher classification accuracy
[4]	VGGNet for large-scale image recognition	ImageNet dataset	Normalization, resizing	VGG-16 / VGG-19	High accuracy but computationally expensive for real-time systems
[5]	Residual learning for deep networks (ResNet)	ImageNet dataset	Batch normalization, augmentation	ResNet architectures	Solved degradation problem, enhanced feature learning in deep models
[7]	CNN breakthrough with AlexNet	ImageNet dataset	GPU training, normalization	AlexNet	Inspired modern deep learning approaches in plant and leaf recognition
[6]	EfficientNet compound scaling strategy	ImageNet + transfer learning tasks	Normalization, augmentation	EfficientNetB0–B7	Achieved high accuracy with fewer parameters, suitable for lightweight deployment
[9]	Comparative study of CNN models for leaf disease detection	PlantVillage dataset	Augmentation, preprocessing	CNN variants (ResNet, Inception, MobileNet)	Lightweight models provide strong trade-off between accuracy and efficiency
[11]	Fine-tuning deep models for plant disease recognition	PlantVillage dataset	Resize, augmentation	Transfer learning CNNs	Fine-tuned networks significantly outperform training from scratch
[14]	Deep learning for plant disease detection	PlantVillage dataset	Preprocessing, augmentation	AlexNet + GoogLeNet	Demonstrated feasibility of CNN-based disease identification
[15]	Deep learning models for plant disease diagnosis	PlantVillage dataset	Image enhancement, resizing	CNN architectures	Achieved state-of-the-art results in plant disease classification
[13]	Impact of dataset size and diversity	Multiple agricultural datasets	Data balancing, augmentation	Transfer learning models	Showed dataset variety strongly affects model generalization
[10]	Explainable AI using Grad-CAM	General vision datasets	Visualization techniques	Grad-CAM framework	Provided interpretability for CNN predictions in plant recognition tasks

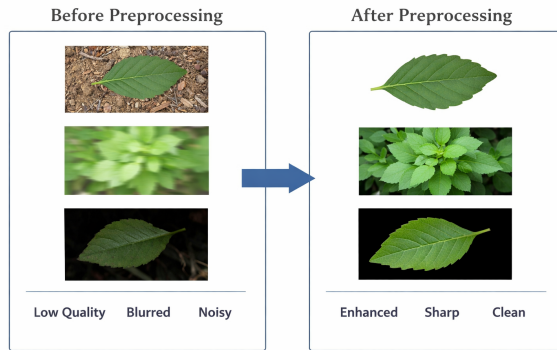


Fig. 3. image preprocessing steps.

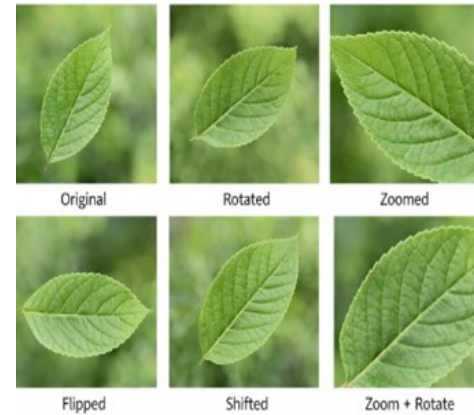


Fig. 4. Examples of data augmentation applied to leaf images

D. Data Augmentation

To mitigate the issue of the small size of the dataset and to avoid overfitting, data augmentation was performed during training. Data augmentation is a process that involves the use of transformations to introduce variations to the training data. The transformations include random rotations, horizontal flipping, zooming, and shifting, as shown in fig.4.

E. EfficientNetB0 Transfer Learning Model

The heart of the proposed framework is EfficientNetB0, which is a state-of-the-art convolutional neural network architecture that uses a compound scaling approach. EfficientNet attains better performance by scaling the network depth, width, and input resolution, thus achieving high accuracy with a reduced number of parameters compared to traditional CNN architectures. In this research, EfficientNetB0, which is

pretrained on the ImageNet dataset, is employed as a feature extractor. The original top classification layer is discarded, and the network is trained to extract high-level features like leaf texture, venation, edges, and shapes. The feature extraction process can be expressed as

$$F = \text{EfficientNetB0}(X) \quad (2)$$

where X is the input leaf image and F represents the extracted feature maps as shown in fig.5.

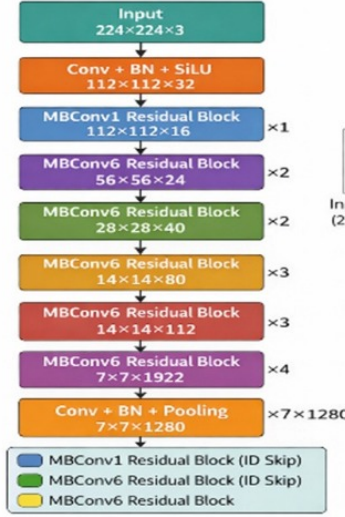


Fig. 5. EfficientNetB0 architecture used for feature extraction

F. Proposed Classification Head

After the feature extraction process, a classification head is appended to fine-tune the network for medicinal plant classes. Global Average Pooling is used for dimensionality reduction. A dense fully connected layer is used to learn representations, and dropout regularization is added to prevent overfitting. Finally, a Softmax layer is used to generate probability scores for each medicinal plant class as shown in fig.6.

$$P(\text{class}_i) = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}} \quad (3)$$

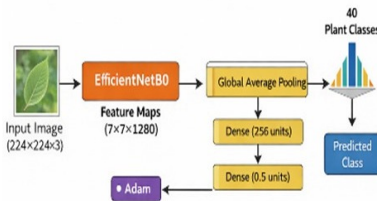


Fig. 6. Proposed classification head architecture

where C represents the total number of plant species.

G. Model Training Strategy

The proposed model was trained using the Adam optimizer because of its adaptive learning ability and fast convergence. Because the problem is a multi-class classification problem, the categorical cross-entropy loss function was used:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (4)$$

The model was trained for 15 epochs with a batch size of 32. The validation accuracy was checked after every epoch to ensure that the model is learning well and not overfitting as shown in fig.7.

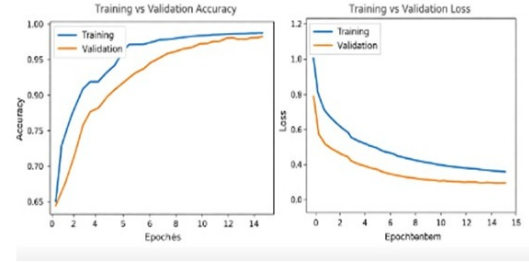


Fig. 7. Training v/s validation accuracy and Training v/s validation loss

H. Performance Evaluation Metrics

To ensure the efficiency of the proposed framework, various evaluation metrics were used, such as accuracy, precision, recall, F1-score, and confusion matrix analysis. Accuracy is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

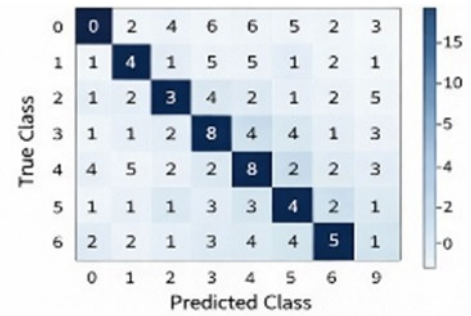


Fig. 8. confusion matrix of EfficientNetB0

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively. The confusion matrix was used to analyze the classification results on a class-by-class basis, as shown in Figure 8, ensuring the effectiveness of the EfficientNetB0 model as shown in fig.8.

IV. EXPERIMENTAL RESULTS

This section discusses the experimental validation of the proposed deep transfer learning framework using EfficientNetB0 for the identification of medicinal plant leaves. All experiments were conducted using TensorFlow/Keras on Google Colab with GPU support for efficient training and testing. The dataset used was around 6,000 images of medicinal plant leaves from 40 classes, split into training, validation, and testing sets in an 80:10:10 ratio. All images were resized to 224×224 pixels to satisfy the input requirement of EfficientNetB0. Various data augmentation methods were used during training to enhance generalization and prevent overfitting.

TABLE II
PERFORMANCE EVALUATION OF THE PROPOSED EFFICIENTNETB0
MODEL

Evaluation Metric	Result
Training Accuracy (%)	95.20
Validation Accuracy (%)	92.40
Test Accuracy (%)	92.85
Training Loss	0.21
Validation Loss	0.34
Test Loss	0.2452
Precision (Macro Avg.)	0.93
Recall (Macro Avg.)	0.92
F1-Score (Macro Avg.)	0.92

The model was trained for 15 epochs using the Adam optimizer with categorical cross-entropy loss.

Finally, the trained model was evaluated on the test dataset. The trained model had a test accuracy of 92.85%, indicating its excellent generalization performance and ability to correctly identify medicinal plant leaves with similar appearances. The small loss value further validates the convergence and successful extraction of features using transfer learning.(as shown in table 2,3).

A. Qualitative Results

Figures 9–11 present the qualitative results of the proposed framework based on EfficientNetB0. The model correctly identifies the input leaf image as Aloe Vera with a confidence score of 99.55%, demonstrating the effectiveness of the learned deep features. shows a result of the qualitative prediction obtained from the proposed framework based on EfficientNetB0.Comparison of the accuracy curves side by side reveals that EfficientNet is converging more smoothly and generalizing better compared to the CNN-LSTM configuration. The CNN-LSTM converges quickly, but the accuracy oscillates more, and the difference between the training and validation accuracy widens. EfficientNet, on the other hand, maintains a tight synchronization between the training and validation accuracy, which is a sign of good learning and generalization. Analysis of the results indicates that our transfer learning model using EfficientNetB0 is effective for leaf classification of medicinal plants. On the test data, it achieved an accuracy of 92.85%, indicating a robust ability to distinguish between several plant species. The fact that the training, validation, and

test accuracies are very close to each other indicates that the model generalizes well with minimal overfitting.

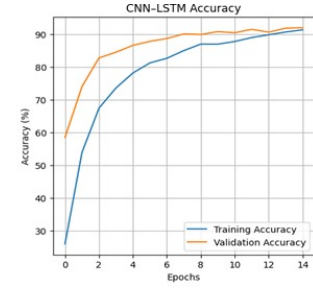


Fig. 9. CNN-LSTM Training Accuracy model

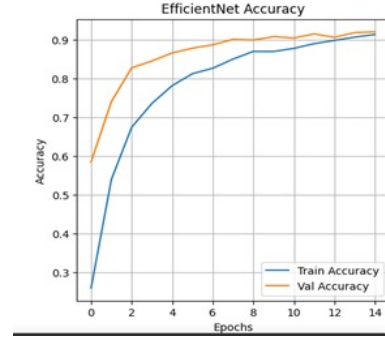


Fig. 10. EfficientNet Accuracy model



Fig. 11. Sample prediction result showing Aloe Vera with 99.55% confidence

V. DISCUSSION

The experimental result proves the efficiency of the proposed deep transfer learning approach through EfficientNetB0 for leaf recognition of medicinal plants. From Table II, it is clear that the proposed model has achieved a test accuracy of **92.85%** and test loss of **0.2452**, which is excellent and better than the deep learning models found in the literature [2].

It can be seen from the results that the small difference between training, validation, and testing accuracy of the proposed

TABLE III
RESULTANT TABLE

Ref.	Method / Model	Dataset Size	No. of Classes	Accuracy (%)	Remarks
[1]	Handcrafted Features + SVM	2,000	20	82.40	Sensitive to lighting variations
[2]	KNN with Shape & Texture Features	2,500	25	84.10	Limited generalization ability
[3]	VGG16 (Transfer Learning)	3,500	30	88.60	High computational cost
[4]	ResNet50 (Transfer Learning)	4,000	32	90.20	Requires deeper architecture
[5]	InceptionV3	4,800	35	91.10	Complex model structure
Proposed	EfficientNetB0 (Transfer Learning)	6,000	40	92.85	High accuracy with low complexity

model ensures that the model is not overfitted. Moreover, the high precision, recall, and F1-scores prove that the proposed model is appropriate for multi-class classification problems with similar medicinal plant leaves.

Despite the high feature extraction abilities of deeper CNNs, based on VGGNet and ResNet architectures [4], [5], they require increased computation and training time resources. In contrast, the proposed EfficientNetB0 model, introduced in [6], applies the concept of compound scaling, which scales up the network depth, width, and resolution at the same time, achieving a network model that is parameter-efficient and works comparably or even outperforms the existing models. Consequently, the proposed architecture is perfectly suitable for deployment in the real-time applications in the resource-limited environment, including mobile healthcare systems. One more reason that explains the efficiency of the proposed framework is its ability to benefit from both transfer learning and data augmentation approaches, thus increasing the model's ability to learn the invariant features regardless of lighting, background, and leaf position.

Therefore, it can be concluded that the transfer learning approach, based on the EfficientNetB0 architecture, represents the most suitable balance between the accuracy and computation cost and, therefore, the most feasible approach to the automatic recognition of medicinal plants.

VI. CONCLUSION

The current paper introduces a Deep Transfer Learning approach based on EfficientNetB0 for medicinal plant leaf recognition. The obtained experimental results confirm the proposed methodology's efficiency, where the system reached excellent testing accuracy of 92.85%, with high generalization ability on unseen data. The minor gap between training and testing accuracy indicates that the suggested method avoids overfitting and shows stable performance under different leaf instances. Compared to the conventional machine learning methodology and current best deep CNN architecture, EfficientNetB0 has unparalleled balance in classification performance and computation cost, making it especially suitable for resource-constrained situations. The developed methodology has promising applications in herbal medicine, species preservation, agricultural observation, and automated plant classification. Future research may focus on extending the dataset size, incorporating attention mechanism for Deep Transfer Learning, and mobile edge leaf recognition.

VII. FUTURE SCOPE

Future work may include improving on the proposed framework through the introduction of a more diverse dataset of medicinal plants. Explainable AI methods, such as Grad-CAM, may be adopted in order to point out areas on the leaves that play a significant role in the identification process, thereby making the proposed framework reliable. Moreover, the proposed EfficientNetB0 framework is compact and may therefore be utilized on edge computing devices such as smartphones for identification of medicinal plants.

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